

Tanzimul Alam Fahim

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Summary

PhD applicant in engineering with research experience in robotics, autonomous systems, & computational modeling. Experienced in Python, ROS, LiDAR mapping, and machine learning for analyzing system behavior and decision-making. Developed autonomous UAV/UGV systems and large-scale cyber threat intelligence knowledge graphs. Interested in computational intelligence, digital twins, and forensic reconstruction of system behavior.

Research Interests: Computational Intelligence, Robotics and Autonomous Systems, Digital Forensics and Behavioral Reconstruction, Machine Learning and Knowledge Graphs, Digital Twins and Intelligent Infrastructure.

Skills

- **Programming Languages:** Python, Java, JavaScript, C++
- **Technologies:** Git, GitHub, VSCODE, Jupyter Notebook, LiDAR/RTAB-Map, MATLAB, Figma/XD, Featherweight iFIP, Mission Planner, PX4/ARDU PILOT, ROS, Autonomous Navigation, Solidworks, Blender, Autodesk fusion, Gazebo, Mediapipe Holistic
- **Data Science & ML:** TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, LangChain, Machine Learning, Computer Vision
- **Security Tools:** Splunk, Cisco AMP, Cisco Umbrella, Cisco XDR, Cisco Threat Response, Cisco Talos, Microsoft, Anomali
- **Tools & Others:** AWS, Docker (Basic), ServiceNow, Microsoft Azure (Basic), Jenkins, Jira, SIEM, Maya, Photoshop, Premier Pro

Research Projects

SAKURA: Knowledge Graph Generation and Analysis

Aug 2023 - Present

- Developed a platform to generate synthetic STIX 2.1-compliant cyber knowledge graphs for CTI analysis.
- Leveraged LangChain with GPT-4o Turbo for realistic, contextually accurate data generation, simulating complex cyber threat relationships.
- Built a dynamic Single Page Application (SPA) with Angular and Tailwind CSS for user-friendly graph generation and visualization.

Link Prediction in Knowledge Graphs (KG)

Sep 2023 - Present

- Conducting comprehensive evaluation of multiple link prediction models with SAKURA dataset for KG applications.
- Implementing and testing existing models, analyzing performance based on accuracy and computational efficiency.
- Performing comparative analysis between existing models and the SAKURA project's approach.

PhishNet: Intelligent Phishing Detection & Threat Analysis System

2026

- Developed a Python-based ML pipeline to detect phishing attacks by modeling structural & behavioral patterns in web and URL data

SkeleVision: AI-Based Human Motion Analysis & Learning System

2026

- Built a computer vision system using MediaPipe & machine learning to analyze human skeletal motion and model behavior patterns for intelligent interaction and learning.

Golapi Care for Wearables

2023

- An ML-based approach to identify Mild Cognitive Impairment among the elderly using data from day-to-day wearable devices.

HealCare AI Chatbot

2023

- Built an AI-powered healthcare chatbot using Django with GPT-4 API fine-tuned for health queries & over-the-counter medications

Autonomous Rescue Drone | Unmanned Aerial Vehicle (UAV)

2022

- Designed for tasks like fire rescue mission, and delivering emergency supplies (like medicine, water) inaccessible areas.

Autonomous Rescue Robot | Unmanned Ground Vehicle (UGV)

2022

- Developed for navigating difficult terrain, useful in scenarios like search and rescue after disasters, automated logistics within a warehouse, or as a research platform for autonomous navigation algorithms.

Mongol Tori Rover | Unmanned Ground Vehicle (UGV)

2022

- It is designed to excel in four core missions: extreme terrain traversal, autonomous navigation, scientific soil analysis, and equipment servicing. The rover acts a platform for hands-on engineering education and international competition.

Professional Experience

UNCP Division of Information Technology, Network & Security Department

Jan 2024 - Sep 2025

Security Operations Center (SOC) Analyst

Pembroke, NC

- Proactively investigated and conducted vulnerability assessments using Splunk, Microsoft Defender, Cisco AMP, Umbrella, and PhishER; developed Python scripts and applied machine learning-based anomaly detection to automate threat identification, reducing incident response time by 30%.

UNCP Department of Mathematics & Computer Science

Aug 2024 - May 2025

Teaching Assistant (TA) & Web Information Coordinator (WIC)

Pembroke, NC

- Assisted in Applied Cryptography course through syllabus development, assignment design, and student support, managing version control via GitHub to streamline collaboration.
- Creating and distributing marketing emails for the department to share cybersecurity resources, news, and opportunities. This initiative encourages involvement in cybersecurity activities, promoting events, and fostering a collaborative environment.

- Keeping the department's website updated to ensure that information is current and relevant for students and visitors.

UNCP Department of Geology

May 2024 - May 2025

Data Science Research Assistant

Pembroke, NC

- Built Python pipelines processing 100k+ PFAS well-water records; cut manual cleaning time by 60% via automated validation, Haversine matching, and data visualization in Jupyter Notebook.
- Applied Principal Component Analysis (PCA) with Scikit-learn for dimensionality reduction, contributing to downstream model performance and enabling deeper data insights.
- Engineered robust data integration pipelines with Python, merging datasets and ensuring data consistency and integrity through validation techniques.

UNCP Cyber Defense Education Center

Jan 2024 - May 2025

Research Assistant

Pembroke, NC

- Researched and enhanced Mobile Driver's License (mDL) architecture, focusing on security, privacy, interoperability, and user experience.
- Analyzed 100+ studies on digital identity, cryptographic protocols, and mobile security to identify research gaps and propose innovations.
- Investigated scalable, secure mDL architectures—including blockchain, biometric authentication, and communication protocols (NFC, Bluetooth, QR codes)—and evaluated performance through MATLAB simulations.

UNCP Housing & Residence Life

Jan 2024 - Apr 2025

Resident Assistant

Pembroke, NC

- Managed field & administrative responsibilities and ensured a harmonious living environment of 577 residents

Laboratory for Analytic Sciences and National Security Agency

Dec 2023 - Aug 2024

Research and ML Intern

Raleigh, NC

- Led a team of 5 in developing a large-scale synthetic cyber threat intelligence (CTI) software, generating a knowledge graph with 100,000+ nodes
- Developed a Django-based web application for user-defined attack scenario and entity selection
- Designed graph-based computational models representing adversarial behavior patterns
- Engineered Generative Adversarial Networks (GANs) using PyTorch for synthetic data generation
- Integrated LangChain & GPT-4 to auto-generate STIX 2.1 objects with 90% validity; presented results at LAS Research Symposium

Laboratory for Space Systems Engineering & Technology

Feb 2021 - Jul 2022

Research Assistant

Dhaka, Bangladesh

- Led development of autonomous aerial and ground robotic systems using ROS, LiDAR, and sensor fusion
- Designed autonomous navigation, mapping, and perception pipelines
- Led team to European Robotics League 2022 Finals — first Asian Team, Perseverance Award Winner
- Developed rover subsystems for autonomous terrain navigation and exploration
- Applied computational modeling and system integration for autonomous control

Education

University Of North Carolina at Pembroke

2025

Bachelor of Science, Cybersecurity

- **Achievements:** Honors Scholar Fellowship, Chancellor's Honor List, National Honors Society, Pembroke Scholar 2023, 2024 & 2025, UNCP Maynor Honors College Student Council Chairman (2024-2025), UNCP SGA Electoral Liaison for 77th Election.

Certifications

- **Google Cybersecurity Professional:** Google Professional certification validating foundational skills in network security, threat detection, and incident response.
- **ISO 27001:2022 ISMS Lead Auditor:** Mastermind Certifies the ability to audit an Information Security Management System against the ISO 27001:2022 standard.
- **Certified in Cybersecurity:** International Information Systems Security Certification Consortium (ISC)2 Foundational cybersecurity certification from (ISC)2, demonstrating core security knowledge and principles.
- **MATLAB Onramp:** MathWorks Demonstrates proficiency in the core functionality of the MATLAB programming and computational environment.

Leadership Experience

AI@UNCP | UNC Pembroke

Aug 2023 - Dec 2025

Vice President

Organized monthly programming competitions in collaboration with ACM Chapter at UNCP.

HackUNCP Hackathon | UNC Pembroke

Aug 2024 - Dec 2025

Organizer

Coordinating logistics, sponsorships management, industry engagement and participant engagement for the hackathon.